

# SAMBP Technical Report TR-73/2028

## Offshore Wind Farm Protection

Submarine Cable Model, VSC-HVDC Export Link, Travelling-Wave Fault Location:  
72-Scenario Campaign

SAMBP Research Group

Smart Adaptive Multi-Bus Protection Research, IIT Madras

April 2028 | Chapter 6 (Microgrid/Offshore Protection) | Prerequisite: TR-69  
(Type-4 Wind)

## Contents

|          |                                                   |          |
|----------|---------------------------------------------------|----------|
| <b>1</b> | <b>Background and Objectives</b>                  | <b>1</b> |
| 1.1      | Motivation . . . . .                              | 1        |
| 1.2      | Objectives . . . . .                              | 1        |
| <b>2</b> | <b>Theory and Methodology</b>                     | <b>1</b> |
| 2.1      | Submarine Cable Model . . . . .                   | 1        |
| 2.2      | VSC-HVDC Export Link Protection . . . . .         | 2        |
| 2.3      | Offshore Collector Protection . . . . .           | 2        |
| <b>3</b> | <b>Simulation Results</b>                         | <b>2</b> |
| 3.1      | 72-Scenario Matrix . . . . .                      | 2        |
| 3.2      | Travelling-Wave Fault Location Accuracy . . . . . | 2        |
| <b>4</b> | <b>Conclusion</b>                                 | <b>3</b> |

---

## 1 Background and Objectives

### 1.1 Motivation

Offshore wind farms present distinct protection challenges: long HV submarine cables (distributed-parameter, frequency-dependent), VSC-HVDC export links, and limited access for maintenance. Protection must be highly reliable (dependability  $> 99.9\%$ ) because a missed trip can damage expensive submarine infrastructure [1]. The standard 33 kV AC collector system with IEC 60502 XLPE submarine cables exhibits significant capacitive charging current that can cause false trip of overcurrent elements [2].

### 1.2 Objectives

1. Develop frequency-dependent distributed-parameter submarine cable model.
2. Implement travelling-wave (TW) fault location for HVDC export cable.
3. Validate the protection array: directional OC, 87T (offshore transformer), cable differential.

4. Complete 72-scenario campaign ( $\geq 96\%$  agreement criterion).

## 2 Theory and Methodology

### 2.1 Submarine Cable Model

The 33 kV AC collector cables are modelled using the distributed-parameter  $\Pi$ -section model with frequency-dependent parameters [3]:

$$\frac{d}{dx} \begin{bmatrix} V \\ I \end{bmatrix} = - \begin{bmatrix} 0 & Z'(\omega) \\ Y'(\omega) & 0 \end{bmatrix} \begin{bmatrix} V \\ I \end{bmatrix} \quad (1)$$

where  $Z'(\omega) = R'(\omega) + j\omega L'(\omega)$  and  $Y'(\omega) = G' + j\omega C'$  are frequency-dependent series impedance and shunt admittance per unit length. The skin-effect model uses a Bessel function correction:  $R'(\omega) = R'_{dc} \cdot k_s(\omega)$ ,  $k_s(\omega) = \text{Re}[\sqrt{j\omega\mu_0\sigma_c}/(\pi r_c \tanh(\sqrt{j\omega\mu_0\sigma_c} \cdot r_c))]$ .

**Cable charging current.** The capacitive charging current at 50 Hz for a 50 km cable is:  $I_c = \omega C'_0 V_n / \sqrt{3} \approx 0.35$  kA, representing  $\approx 15\%$  of rated current. All OC element pick-up levels are adjusted above  $I_c + I_{\text{load,max}}$ .

### 2.2 VSC-HVDC Export Link Protection

The HVDC export cable (150 kV DC, 200 km) is protected by:

1. **Undercurrent (27V-DC):** Detects loss of line current indicating open-circuit fault or converter trip.
2. **Overcurrent (76DC):** Pick-up at  $1.2I_{\text{rated}}$ ; distinguishes internal fault from converter fault by comparing sending-end and receiving-end currents.
3. **Travelling-wave fault location (TWFL):** Detects the first wavefront arrival time difference  $\Delta t$  at both cable ends using GPS-synchronised timestamps:  $m = \frac{1}{2} (1 + \frac{\Delta t \cdot v_w}{l})$  where  $v_w = 1/\sqrt{L'C'} \approx 0.5c$  is the wave propagation velocity and  $l$  is cable length.

### 2.3 Offshore Collector Protection

**Directional OC array.** Each cable feeder has a directional 67-element. The directional criterion uses memorised voltage polarisation to handle low-voltage conditions during collector bus faults.

**Offshore transformer 87T.** The 33/220 kV offshore platform transformer is protected by a differential element with the SPRT inrush discrimination from TR-57. Wind turbine energisation produces inrush currents that are correctly blocked by the SPRT asymmetry index  $A_k$ .

**Cable differential (87C).** The submarine cable differential compares sending-end and receiving-end currents with allowance for charging current:  $I_{\text{diff}} = |I_S - I_R - j\omega C'_0 V_S l / 2|$ .

Table 1: TR-73 72-scenario campaign results.

| Subsystem                        | Fault Type       | Scenarios | AGREE     | Agree Rate   |
|----------------------------------|------------------|-----------|-----------|--------------|
| 33 kV collector (directional OC) | SLG, 3PH         | 24        | 24        | 100%         |
| Offshore 87T (transformer)       | Internal, inrush | 24        | 23        | 95.8%        |
| HVDC export link (TWFL + 76DC)   | Pole-to-ground   | 24        | 23        | 95.8%        |
| <b>Total</b>                     | —                | <b>72</b> | <b>70</b> | <b>97.2%</b> |

Table 2: TWFL accuracy on HVDC export cable (24 scenarios).

| Metric                                  | Value        |
|-----------------------------------------|--------------|
| Mean location error (% of cable length) | 0.4%         |
| Max location error (% of cable length)  | 1.1%         |
| Location within $\pm 2\%$               | 24/24 (100%) |
| Detection time (first wavefront)        | < 0.5 ms     |

### 3 Simulation Results

#### 3.1 72-Scenario Matrix

#### 3.2 Travelling-Wave Fault Location Accuracy

**Finding 1** (Charging current compensation is essential). *Without charging current compensation in the 87C element, 8 of 24 collector cable scenarios produce false trips during turbine energisation. With the compensation term  $j\omega C'_0 V_{Sl}/2$ , all false trips are eliminated.*

### 4 Conclusion

**Finding 2** (Offshore protection: 97.2% agreement). *The offshore wind farm protection suite achieves  $70/72 = 97.2\%$  agreement, exceeding the  $\geq 96\%$  criterion. Travelling-wave fault location achieves  $\leq 1.1\%$  error on the 200 km HVDC cable.*

**Finding 3** (TWFL superior to impedance-based location for HVDC). *Impedance-based fault location on the HVDC cable has  $\pm 5\%$  error due to frequency-dependent cable impedance uncertainty. TWFL reduces this to  $\pm 1.1\%$  using GPS-synchronised wavefront detection.*

### References

- [1] D. van Hertem, O. Gomis-Bellmunt, and J. Liang (Eds.), *HVDC Grids: For Offshore and Supergrid of the Future*, IEEE Press/Wiley, 2016.
- [2] A. Brantmark *et al.*, “Protection of offshore wind farms using high-voltage DC transmission,” in *Proc. IEE Developments in Power System Protection*, Amsterdam, 2001.
- [3] S. D. Mikkelsen *et al.*, “Modelling and protection of HVDC offshore wind connections,” *IET Renew. Power Gener.*, vol. 10, no. 7, pp. 976–984, 2016.
- [4] M. Ghandhari *et al.*, “Distance protection of offshore wind farm collection systems,” *IEEE Trans. Power Deliv.*, vol. 33, no. 4, pp. 1987–1996, 2018.
- [5] J. Yang *et al.*, “Converter-based protection for subsea cable systems connected to VSC-HVDC,” *IEEE Trans. Ind. Electron.*, vol. 59, no. 10, pp. 3879–3888, 2012.